

Ajay Kumar Saini

Email: nabalsaini231@gmail.com

Phone: +91-8107469345

GitHub: github.com/ajaykumarsaini231

LinkedIn: linkedin.com/in/ajay-kumar-saini

EDUCATION

Indian Institute of Technology (IIT) Patna

B.Tech in Metallurgical and Materials Engineering | **CPI: 7.82**

Patna, Bihar

August 2023 – Present

Minor Focus: Algorithms, Data Structures, Database Systems, Machine Learning

Relevant Coursework: DSA, DBMS, Operating Systems, Linear Algebra, Probability

Bhagawati Sr. Sec. School, Gangapur City

Class XII (91.4%)

Rajasthan

April 2022 – April 2022

EXPERIENCE

Indian Institute of Technology Jodhpur

Research Intern, Dept. of Metallurgical and Materials Engineering

Jodhpur, India

May 2025 – July 2025

- Analyzed solidification cracking susceptibility and phase transformations using Kou's model.
- Validated numerical predictions through correlation with XRD and SEM microstructural data.

Anwesha – Cultural Fest Website

Technical Lead / Head (Web Development)

anwesha.iitp.ac.in

July 2024 – Current

- Led development of the official platform supporting **3,000+ active registrations** with zero downtime.
- Improved page load performance by **40%** using React lazy loading and optimized asset delivery.

E-Cell – Entrepreneurship Cell Website

Technical Lead / Head (Web Development)

ecell.iitp.ac.in

July 2024 – Current

- Built and maintained a portal managing data for **50+ startups** and **500+ attendees**.
- Optimized database queries to ensure **99.9% data integrity** during concurrent sign-ups and ticketing.

PROJECTS

Dreamora – Full-Stack E-Commerce Platform

Next.js, Express.js, PostgreSQL, Tailwind CSS

Live Demo | GitHub

- Designed REST APIs with indexed PostgreSQL schemas to support concurrency-safe transactions.
- Reduced API latency by **30%** using caching and query optimization.
- Implemented secure authentication and authorization using JWT and RBAC, handling **100+ concurrent sessions**.

Questivo – AI-Powered Mock Test Platform

React 19, Node.js, Groq AI, Prisma

Live Demo | GitHub

- Developed an AI-based engine generating **1,000+ unique questions** with contextual relevance.
- Integrated embedding-based similarity search to improve prompt relevance and retrieval accuracy.
- Achieved sub-second inference latency using Groq acceleration for real-time Llama 3.3 responses.

TECHNICAL SKILLS

CS Fundamentals: Data Structures & Algorithms (650+ problems), Backend Systems, API Design, Database Optimization

Languages: C, C++, Python, JavaScript, TypeScript, SQL

Web Technologies: React.js, Next.js, Node.js, Express.js, Tailwind CSS, Prisma, JWT, RBAC

Databases & Tools: PostgreSQL, MongoDB, Git, GitHub, Postman, Vercel, VS Code

Soft Skills: Leadership, Team Coordination, Communication, Public Relations

ACHIEVEMENTS

Solved **650+ DSA problems** across multiple platforms, covering core data structures and algorithms.

Ranked in the **Top 2%** among **1.2M+** candidates in JEE Advanced 2023.

Placed in the **Top 2.44%** of **860,000+** candidates in JEE Mains 2023.

Management Head, Celesta: Coordinated operations for IIT Patna's Techno-Management Fest.